

- The main stages of cell signalling involving ACTH are similar to the main stages of cell signalling by adrenaline.

[5]

- [1]

- (c) The domestic dog, *Canis lupus familiaris*, is a mammal. A dog may develop a condition called Cushing's disease, which is a result of too much cortisol being secreted by the adrenal glands. The most common cause of this is an increased secretion of ACTH because of a tumour in the anterior pituitary gland.

Explain how a tumour develops.

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..... [2]

- (d) One symptom of Cushing's disease in a dog is that the dog will want to drink much more than normal.

Suggest **one other** reason why a dog would want to drink much more than normal.

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..... [1]

[Total: 9]